

Exercise 2

Write a more advanced version of the News page by adding a `news_advanced()` view to `home/views.py`. Each data item in this view is a tuple containing the date and subject of the news. Create a `news_adv.html` template that shows the data in a table. Look up the `{% cycle %}` tag and the `|date` filter in appendix C, and apply them to your table, using `{% cycle %}` to apply striping to the table and `|date` to format the news item's date.

This exercise is mostly about getting more practice with some tags and filters, but to do that, you need a view to render your new template. The advanced version of the News page requires a view containing tuples, where each tuple has a date and a news headline. The following listing defines a view that uses `today`, `yesterday`, and the day before as dates and three bits of news to go with them.

Listing E.9 An advanced version of a news view

```
# RiffMates/home/views.py
from datetime import date, timedelta

from django.http import HttpResponse, JsonResponse
from django.shortcuts import render

def news_advanced(request):
    today = date.today()
    before1 = today - timedelta(days=1)
    before2 = today - timedelta(days=2)

    data = {
        "news": [
            (today, "Advanced news added! Even more exclamation points!!!"),
            (before1, "RiffMates now has a news page!"),
            (before2, "RiffMates has its first web page"),
        ],
    }

    return render(request, "adv_news.html", data)
```

Python's `date` contains the `today()` method that returns today's date, while `timedelta` allows you to do math on date objects.

Get today's date.

Calculate the day before today.

Context data for the template engine containing news items for today and the two prior days

You learned about the `render()` shortcut in chapter 3; it loads and renders a template, returning an `HttpResponse` object with the resulting content.

The following listing shows the new view being registered as a route.

Listing E.10 Register the Advanced News view

```
# RiffMates/RiffMates/urls.py
...
from django.urls import path
from home import views as home_views

urlpatterns = [
    # ...
    path("adv_news/", home_views.news_advanced, name="adv_news"),
]
```